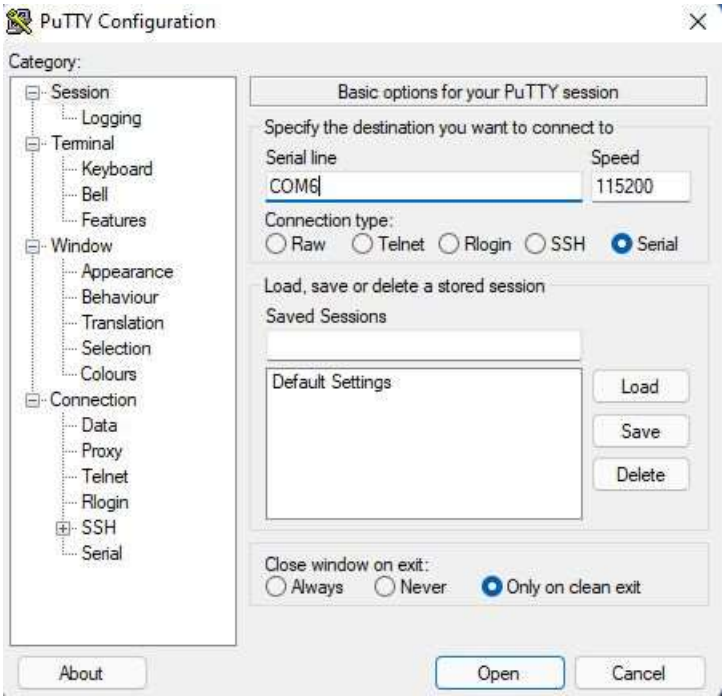
ANTMINER Arcade Linux OS Configuration

- AntMiner S9 ZYNQ-7010 FPGA Board
- USB to Serial Adapter
- Putty Software
- PinballWiz AntMiner-Arcade_v1.2 GitHub Repository Files

USB to Serial Adapter (aliexpress.com)

Putty Software



Serial Communication Setup

Connect Serial Adapter Gnd, RX, TX to AntMiner Gnd, TX, RX pins on serial header as shown below (VCC is not used). Note RX to TX and TX to RX connections below. Connect Serial Adapter to USB Port on Computer. Open Putty Software and setup as shown above:

- Select Serial
- COM# (device manager will show which com port to use)
- Speed = 115200

Click Putty Open Button.
Power On the AntMiner Board.

Putty Should now display the AntMiner Boot process.

```
--- mount angstrom file system
bootargs: noinitrd mem=1008M console=ttyPS0,115200 root=ubi0:rootfs ubi.mtd=
otfstype=ubifs rw rootwait
--- upgrade marker upgrade_buf[0] = 0xff ---
--- upgrade marker upgrade_buf[1] = 0xff ---
--- upgrade marker upgrade_buf[2] = 0xff ---
--- upgrade marker upgrade_buf[3] = 0xff ---
--- upgrade marker upgrade_buf[4] = 0xff ---
--- upgrade marker upgrade_buf[5] = 0xff ---
--- upgrade marker upgrade_buf[6] = 0xff ---
--- upgrade marker upgrade_buf[7] = 0xff ---
--- upgrade marker upgrade_buf[8] = 0xff ---
--- upgrade marker upgrade_buf[9] = 0xff ---
--- upgrade marker upgrade_buf[10] = 0xff ---
--- upgrade marker upgrade_buf[11] = 0xff ---
--- upgrade marker upgrade_buf[12] = 0xff ---
--- upgrade marker upgrade_buf[13] = 0xff ---
--- upgrade marker upgrade_buf[14] = 0xff ---
--- upgrade marker upgrade_buf[15] = 0xff ---
bootargs: noinitrd mem=1008M console=ttyPS0,115200 root=ubi0:rootfs ubi.mtd=
```

ANTMINER Arcade Linux OS Configuration

- Login to AntMiner first by trying login as root with password = admin
- If above login does not work continue with below procedure to reset password.
- Interrupt the normal boot process by pressing any key.
- The Putty Console Window should then show the uboot prompt below.
- The root password can now be changed. (see commands below).

```
--- mount angstrom file system
bootargs: noinitrd mem=1008M console=ttyPS0,115200 root=ubi0:rootfs ubi.mtd=
otfstype=ubifs rw rootwait
--- upgrade marker upgrade_buf[0] = 0xff ---
--- upgrade marker upgrade_buf[1] = 0xff ---
--- upgrade marker upgrade_buf[2] = 0xff ---
--- upgrade marker upgrade_buf[3] = 0xff ---
--- upgrade marker upgrade_buf[4] = 0xff ---
--- upgrade marker upgrade_buf[5] = 0xff ---
--- upgrade marker upgrade_buf[6] = 0xff ---
--- upgrade marker upgrade_buf[7] = 0xff ---
--- upgrade marker upgrade_buf[8] = 0xff ---
--- upgrade marker upgrade_buf[9] = 0xff ---
--- upgrade marker upgrade_buf[10] = 0xff ---
--- upgrade marker upgrade_buf[11] = 0xff ---
--- upgrade marker upgrade_buf[12] = 0xff ---
--- upgrade marker upgrade_buf[13] = 0xff ---
--- upgrade marker upgrade_buf[14] = 0xff ---
--- upgrade marker upgrade_buf[15] = 0xff ---
bootargs: noinitrd mem=1008M console=ttyPS0,115200 root=ubi0:rootfs ubi.mtd=
otfstype=ubifs rw rootwait
Net:   Gem.e000b000
Hit any key to stop autoboot:  0
zynq-uboot>
```

Resetting Root Password

Copy each of these command lines into the Putty Console window - hit enter after each.

```
nand read 0x2000000 0x1100000 ${kernel_size}

nand read 0x3000000 0x1020000 ${devicetree_size}

setenv bootargs 'noinitrd mem=496M console=ttyPS0,115200 root=ubi0:rootfs ubi.mtd=1 rootfstype=ubifs rw rootwait init=/bin/sh'

bootm 0x2000000 - 0x3000000

passwd
```

*** Note if your AntMiner board has 1gb of ram instead of 512mb then use the below setenv line instead of the setenv line above:
setenv bootargs 'noinitrd mem=1008M console=ttyPS0,115200 root=ubi0:rootfs ubi.mtd=1 rootfstype=ubifs rw rootwait init=/bin/sh'

- Choose a root password when prompted and verify password again.
- Restart the AntMiner Board.

Disable Bitcoin Miner Software

- Apply Power to the AntMiner Arcade Board.
- The AntMiner Board should eventually show the below login prompt.
- Login as root using the password set in the previous steps.
- The AntMiner Board should also still be running the Bitcoin Miner Software which will continue to display information in the Putty Console Window.

- To Disable the Bitcoin Miner Software Enter the commands shown below.
- Hit Enter after each command.

```
cd /etc/rcS.d
mv S70bmminer.sh K70bmminer.sh
mv S37bitmainer_setup.sh K37bitmainer_setup.sh
cd /usr/bin
mv bmminer bmminerXX

reboot
```

Remove unwanted Services and add a Static IP Address

- Apply Power to the AntMiner Arcade Board.
- The AntMiner Board should eventually show the below login prompt.
- Login as root using the password set in the previous steps.
- Enter the following commands as shown below.

```
cd /etc/rcS.d
./S60lighttpd stop
mv S60lighttpd K60lighttpd

mv S50avahi K50avahi

chmod 600 /config/network_conf

vi /config/network.conf

hostname=antMiner
dhcp=false
ipaddress=192.168.0.130
netmask=255.255.255.0
gateway=192.168.0.1

(save with :wq)
```

Copying AntMiner Arcade Files to Linux OS

- Format a 2GB or larger MicroSD Card as FAT32
- Copy the Linux folder from the AntMiner-Arcade_v1.2 Repo to the MicroSD Card.
- Place the MicroSD Card into the AntMiner MicroSD Card Slot.
- Copy the following Commands into the Putty Window hitting Enter after each.

```
mkdir -p /mnt/sdcard

mount -t vfat /dev/mmcblk0p1 /mnt/sdcard

cp /mnt/sdcard/linux/etc/init.d/fpga_loader.sh /etc/init.d/fpga_loader.sh

cp /mnt/sdcard/linux/var/s9_dipread.bit /var/s9_dipread.bit

cp /mnt/sdcard/linux/var/game_loader /var/game_loader

ln -s /etc/init.d/fpga_loader.sh /etc/rcS.d/S96fpga_loader.sh

shutdown -h now
```

- Linux OS Configuration is now complete.
- Remove power from the AntMiner Board.
- See included Documentation on the MicroSD Card Games Setup procedure.

Once a MicroSD Card has been setup and loaded with games, Insert the MicroSD Card into the AntMiner MicroSD Card slot. Set the Dipswitch according to the game you wish to load (see AntMiner Arcade GameMap doc for Dipswitch Settings). Connect a VGA Monitor, Audio Speakers and PS2 Keyboard and apply power to the AntMiner Arcade Board. The Game selected should load automatically.